

CHAPTER 17

Project Management

17.1. Introduction

Management of a project should be treated as if it were a one-time task. By "project" we mean a major undertaking that is unlikely to be repeated in exactly the same way at a future time. Any mistakes made are likely to be costly to rectify, and we are very concerned about performing the job tasks correctly. This is in contrast to managing the production of goods or services in which the process is ongoing. That is, production management is repetitive in nature, whereas project management is more of a single, major job.

Examples of project management might include building a football stadium, administering a large research contract, performing major transplant surgery, establishing a production line, earning a college diploma, or raising a child. All of these projects have some similarities. Each is comprised of many tasks or activities, the number of activities depending on the detail in which we wish to describe the project. The activities of each project have precedence relationships. That is, certain activities cannot be started until others have been completed. Each activity has a duration. Sometimes we can predict with great confidence how long an activity will take; at other times we are very uncertain about our estimates. Finally, the activities of each project must be scheduled in such a way that the project is brought to a successful completion.

Example 17.1

As president of Gadgets, Inc., you have been very successful. Through industrial engineering techniques, you have planned properly, anticipated problems, corrected problems, have a high level of employee morale, and now are faced with excess profits. In order to take advantage of tax write-offs, you have decided to build a new plant. It will be bigger and more modern than your current facility, and all of the arrangements from start to finish of your project seem to be more than you can coordinate in your head.

You have already decided to oversee and coordinate the project. You have decided to separate the project into two phases. The first phase will include obtaining marketing and production factors such as sales forecasts, production levels, storage needs, and so on. This phase will also include plant location and layout. This type of project you have been involved in before, and you need a project management tool in which you can estimate the activity times very closely. The second phase is the construction phase of the project about which you feel much less certain. Here you need a project management tool that allows some flexibility or uncertainty in making time estimates.

In both phases of the project you plan to trade direct costs off against indirect costs to achieve a duration for each phase of the project which provides a minimum total cost. You also want to keep a good cost record of each activity for tax and control purposes. Finally, you desire to balance or level your manpower resource needs over time in both phases of the project.

You recognize that CPM and PERT network activity models offer you the needed techniques for properly planning your expansion. Before launching into your project, however, you take time to review CPM and PERT.

Numerous methods have been devised for project management. One that you worked with in Chapter 7 is the Gantt chart. The Gantt chart shows the time needed for activities, as well as the earliest times when activities can begin. Unfortunately, the Gantt chart is not ideally suited to the management of large projects. The interrelationships of the activities are not shown and it is difficult to determine from a Gantt chart how long the start of a project can be delayed without adversely affecting the project. Also, as the project proceeds and we have deviations from the original plan, adjustments and rescheduling are difficult using a Gantt chart if there are many activities.

In an attempt to overcome some of the shortcomings of the Gantt chart, critical path systems were developed in the late 1950s. We shall look at two of the most popular—critical path method (CPM) and program evaluation and review technique (PERT). CPM and PERT have many similarities. They both show activity duration and indicate the earliest activity start times, as does the Gantt chart. In addition, they clearly show precedence interrelationships between activities, allowing us to determine the most critical activities in the chain of events leading to project completion. Activity modifications and progress updating are relatively easy to do with either CPM or PERT, especially now that computer software is available for use on mainframes and personal computers, taking the drudgery out of this work.

Of course, CPM and PERT are likely to take more time to use than a Gantt

chart, and therefore, they are more costly to use. On small projects the Gantt chart may be preferred, but on large projects critical path systems will almost certainly be required. The choice between CPM, PERT, Gantt charts, and other project management tools must be made by the user and depend on the size and complexity of the project, as well as the software available.

17.2. Project Planning Networks

For analysis by CPM or PERT each project must have three properties:

- (1) The project consists of a set of well-defined activities or tasks which, upon completion, signify the end of the project.
- (2) The activities may be started and stopped independently of each other within a given sequence.
- (3) The activities have precedence relationships and must be performed in the proper order.

Both CPM and PERT utilize a network representation to describe and structure the activities of a project. A critical path network is a graph or picture that shows each task to be performed, its predecessor, and its successors. We can best explain the development of a critical path network by developing one using a simple example.

Suppose that we are in charge of coordinating the installation of a precision rain gauge/wind sensor station. Telemetry equipment will be housed inside a small building to continually record all readings. Not all of the activities needed to install the station can be done at once. Neither need they be done one at a time. In order to determine which activities may be done concurrently, we must specify all precedence relationships. These relationships and our estimates of individual activity durations are given in Table 17.1.

Table 17.1. RAIN GAUGE/WIND SENSOR INSTALLATION DATA

Activity	Description	Predecessor	Duration (Days)
A	Layout sensing station	None	5
B	Procure rain gauge/wind sensor	None	25
C	Procure telemetry equipment	None	18
D	Construct first 50% of building	A	14
E	Construct remainder of building	D	14
F	Install rain gauge/wind sensor	B,D	4
G	Install telemetry equipment	C,D	5
H	Connect telemetry equipment to rain gauge/wind sensor	F,G	3
I	Final test	E,H	1

We can now draw a picture of this information in the form of a network as shown in Figure 17.1. This figure tells us a great deal once we know how to read it. First, let us define the essential characteristics of the graph as follows:

- A circle represents a *node*. Arrows, or activities leaving a node, cannot be started until all activities incoming to the node have been completed. The completion of all activities incoming to a node is considered an *event*, as is the start of the project.
- An arrow represents one of the *activities* of the project. Its length is of no significance.
- - → A dashed arrow represents a *dummy activity*. Dummy activities are used to represent precedence relationships. They are not activities in the real sense and have a duration of zero.

It is common to number the nodes, but the number given to a node is irrelevant and does not indicate precedence or duration. It is also common to indicate each activity with a code letter followed by its duration. The code letter is used to maintain a neat graph instead of trying to write each activity description over its respective arrow.

Having drawn the network and defined its elements, we can see, for example, that installation of the rain gauge/wind sensor (F) cannot begin until they have been procured (B) and the building is 50% completed (D). Clearly, the last half of the building cannot be constructed (E) until the first half is completed (D). In this case, a dummy activity is required because activity F requires both B and D to be completed, while E requires only that D be finished. It is also easy for us to observe the total time required for various paths through the network. For example, the path consisting of activities C, G, H, and I (nodes 1, 5, 6, 7, 8) is estimated to consume 27 days. We can also see that the longest path through the network includes activities A, D, E, and I (nodes, 1, 2, 3, 7, 8) requiring 34 days. This would be called the *critical path*.

We can see that critical path techniques offer a number of advantages for project planning, scheduling, and control, for example:

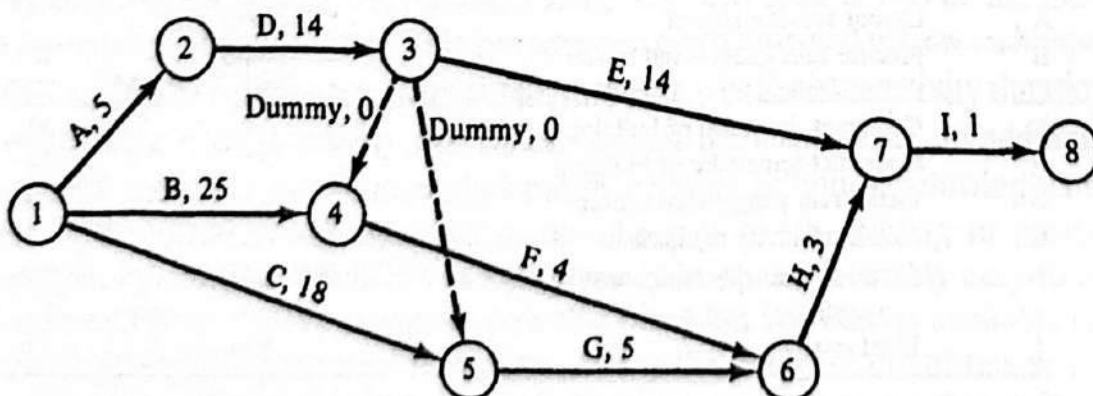


Fig. 17.1. Critical path network of rain gauge/wind sensor installation.

- (1) They encourage a logical approach to project management.
- (2) They encourage long-range and detailed planning.
- (3) They provide a standard method of communicating project plans, schedules, and time performance.
- (4) They focus management attention on the usually small percentage of critical activities.
- (5) They can illustrate the effects of technical and procedural changes on the project schedule.

We shall now look at CPM and PERT in more detail.

17.3. Critical Path Method

Once we have reduced a project to a network of activities and nodes, we are ready to determine the critical path. In the above example we were easily able to identify the critical path as that sequence of activities requiring the longest time to accomplish. As the size of a project grows, however, it becomes almost impossible to identify the critical path through the network. CPM offers a systematic procedure for selecting the critical path. In addition, the amount of slack or free time on noncritical paths may be determined. Knowing the slack on noncritical paths may permit us to trade off manpower and equipment resources from noncritical activities in order to concentrate on and shorten the critical path.

CPM requires that we begin by identifying all project activities, precedence relationships, and times. As an example, we may have a project described as follows:¹

- (1) A, B, and C are concurrent and begin at the start of the project.
- (2) D and E are concurrent and can begin upon the completion of A.
- (3) F may begin upon the completion of B.
- (4) G starts after C is finished.
- (5) H follows D.
- (6) I can begin when E and F are complete.
- (7) Completion of G, H, and I marks the end of the project.

Once we have activities and precedence relationships, we can draw the basic CPM network. This allows *planning*, but in order to do *scheduling* we must have activity time estimates. Some rules for estimating activity times are as follows:

- (1) Obtain time estimates from the best possible source. Estimating must be done by experienced personnel, often the person, group, or subcontractor responsible for accomplishing the activity.

¹Taken from an example used by R. J. Craig and W. C. Turner in teaching extension courses through Virginia Polytechnic Institute and State University, and Oklahoma State University.

- (2) Estimates for activities will be on an independent basis. That is, the estimate for each arrow is made irrespective of estimates made for every other arrow in the diagram.
- (3) The time estimate is based on the usual practical resources available. This means using a regular working crew, a normal working day, and so on.
- (4) Consistent time units must be used throughout.

Suppose that we estimate the activity durations for our example as follows:

A: 8 days	F: 9 days
B: 20 days	G: 10 days
C: 33 days	H: 8 days
D: 18 days	I: 4 days
E: 20 days	

The CPM network for our project can now be illustrated as in Figure 17.2.

We are now ready to analyze the network in detail. To do this, we will use the following nomenclature:

t	= single estimate of mean activity duration
T_E	= earliest event occurrence time
T_L	= latest allowable event occurrence time
ES	= earliest activity start time
EF	= earliest activity finish time
LS	= latest allowable activity start time
LF	= latest allowable activity finish time
S	= total activity slack

Using this notation, we shall now discuss forward and backward passes through the network, total activity slack, and the critical path.

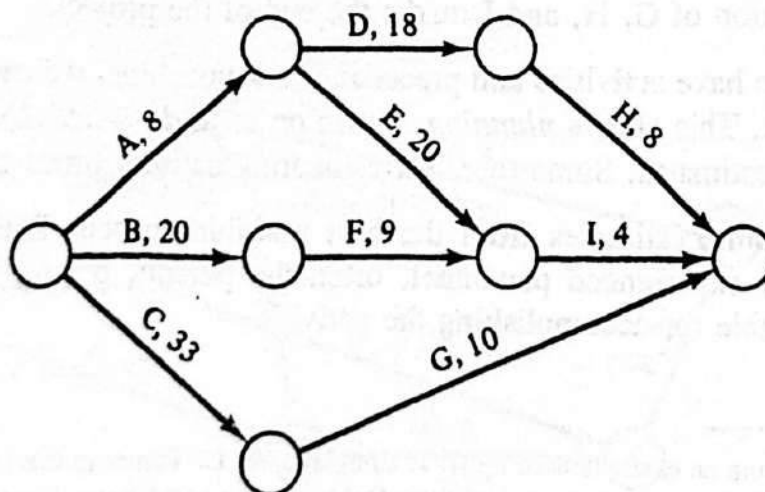


Fig. 17.2. CPM network.

17.3.1 Forward Pass

The computations from a forward pass through the network provide information on the earliest time that an event can be completed. In the forward pass calculations we begin with the first activity (or activities) and proceed along each path of the network observing the following rules:

- (1) The earliest occurrence time of the (single) initial event (node) is taken as zero. That is, $T_E = 0$ for the initial event.
- (2) Each activity begins as soon as its predecessor event (node) occurs. That is, $ES = T_E$ (for its predecessor event). Since the duration of the activity is t , its earliest finish time is $EF = ES + t = T_E + t$.
- (3) The earliest event (node) occurrence time is the largest of the earliest finish times of the activities merging to the event in question. Mathematically, $T_E = \max(EF_1, EF_2, \dots, EF_n)$ for an event with n merging activities.

We can illustrate the forward pass notation pictorially as shown in Figure 17.3. In this figure activity K has a duration of t . It connects events (nodes) i and j . The earliest event occurrence time for node i is T_{Ei} , this also being the earliest start time, ES , for activity K. The earliest finish time for activity K is EF which is t time units after ES . The earliest event time for node j is T_{Ej} , which is the maximum of the earliest finish times of the incoming activities. If only one activity is incoming to node j , as shown, $T_{Ej} = EF$.

Let us now redraw our network by using the forward pass. This is done in Figure 17.4. From this drawing, we can see the earliest start and finish times for each activity, as well as the earliest time at which our events (nodes) may be accomplished.

17.3.2 Backward Pass

The backward pass calculations tell us the latest times at which events can be completed and still meet the final schedule date. The backward pass is identical in nature to the forward pass, except that the backward pass starts with the last node and works toward the starting event. The backward pass rules are as follows:

- (1) The latest allowable occurrence time of the (single) terminal event is set to the same value as the earliest occurrence time that we calculated in the forward pass. That is, $T_L = T_E$ for the last event.

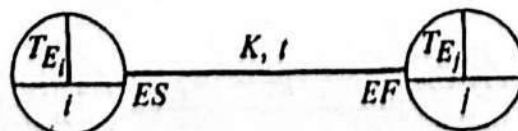


Fig. 17.3. Forward pass notation.

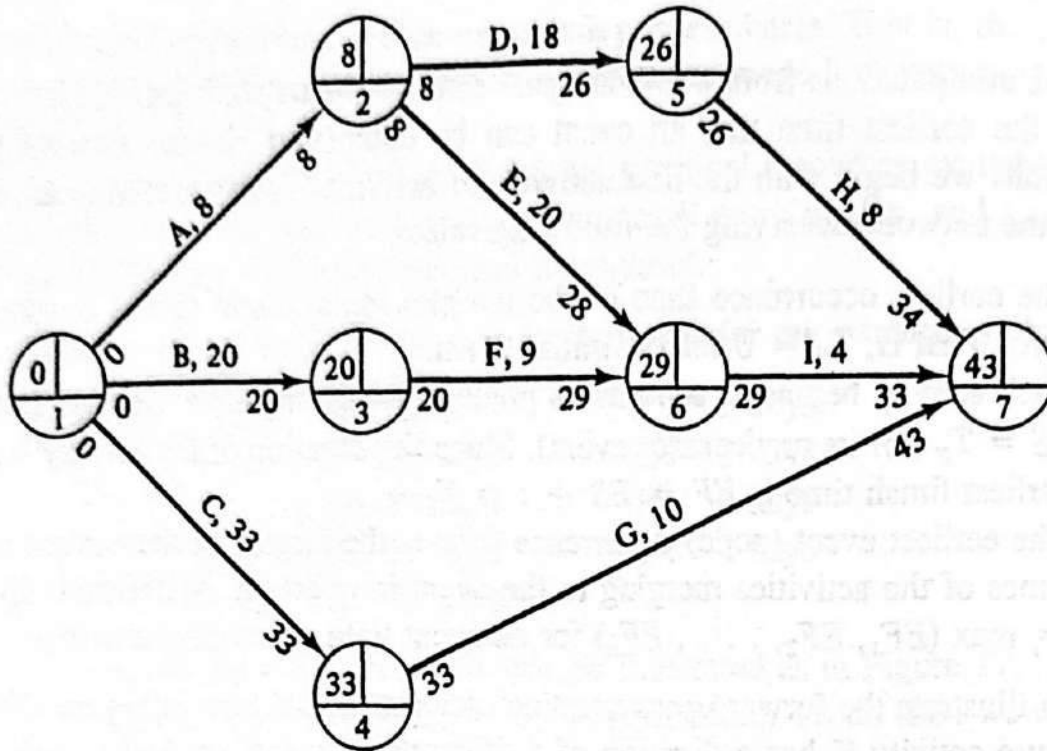


Fig. 17.4. Forward pass calculations.

- (2) The latest allowable finish time for an activity is its successor event's latest allowable occurrence time. For this we have $LF = T_L$ for the successor event. The latest possible activity start time is its latest allowable finish time less the duration of the activity in question. Mathematically, $LS = LF - t = T_L - t$.
- (3) The latest allowable time for an event is the smallest of the latest allowable start times of the activities emanating from the event of interest. That is, $T_L = \min (LS_1, LS_2, \dots, LS_n)$ for an event with n emanating activities.

Let us now add the backward pass notation to that for the forward pass. This is done in Figure 17.5. The latest event time for node j is T_{Lj} , this also being the latest finish time, LF , for activity K . The latest start time for activity K is LS which is t time units before LF . The latest event time for node i is the minimum of the latest start times for the emanating activities. If only one activity leaves node i as shown, $T_{Li} = LS$.

We can now modify Figure 17.4 to give both the forward and backward pass data. This modification is performed in Figure 17.6.

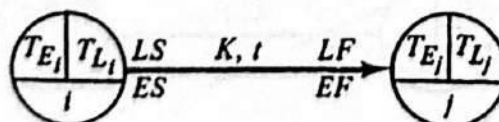


Fig. 17.5. Forward and backward pass notation.

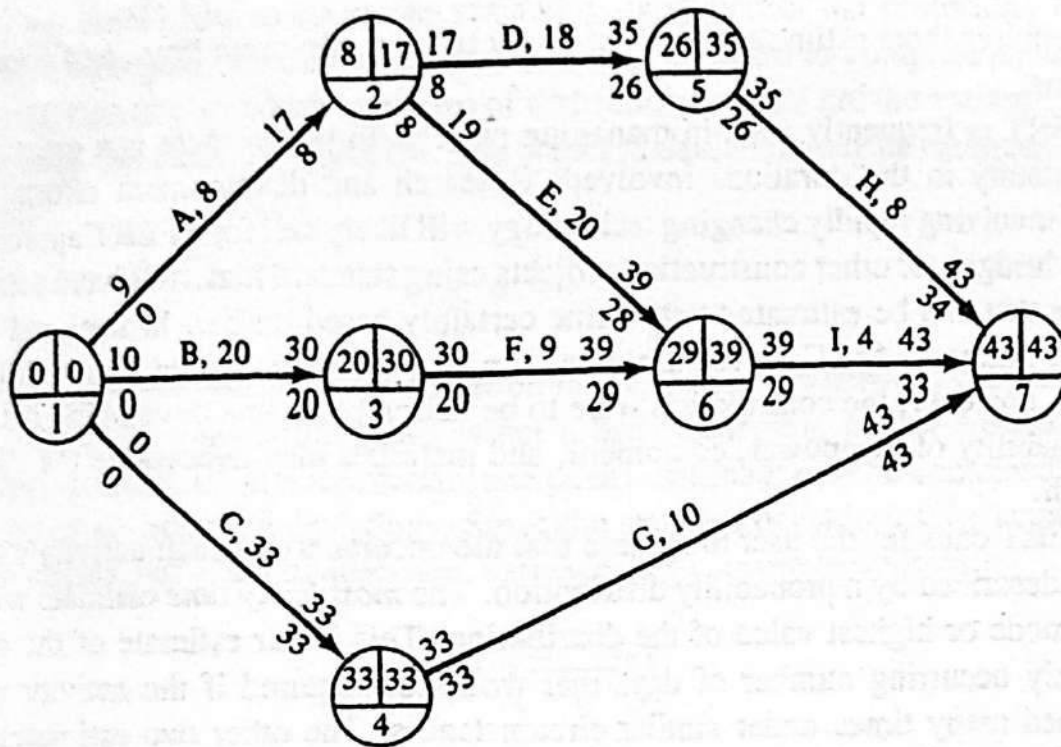


Fig. 17.6. Forward and backward pass calculations.

17.3.3 Total Activity Slack

Total activity slack equals the latest allowable occurrence time of an activity's successor event, less the earliest finish time of the activity of interest. That is, $S = T_L - EF$. The total activity slack tells us the amount of delay we can have for a task without affecting the earliest start of any activity on the critical path. For example, activity E has 11 days of slack. It can be started as early as day 8 or as late as day 19 without extending the project duration beyond 43 days.

17.3.4 Critical Path

The critical path may be identified as the sequence of activities having no slack. We can see that $ES = LS$ and $EF = LF$ for each activity on the critical path, and $T_E = T_L$ for each node. The critical path is one in which a delay in any activity on the path will cause a delay in the project. Our critical path is easily identified as activities C, G, or nodes 1, 4, 7.

17.4. Program Evaluation and Review Technique

PERT was developed in an independent effort at about the same time CPM was established. It utilizes a project network, has a critical path, and maintains the notion of total slack. In what way do they differ? The basic difference is in the way activity durations are estimated. Whereas CPM requires a single estimate for each activity,

PERT requires three estimates—a most likely time, a minimum time, and a maximum time.

PERT is frequently used in managing projects in which there is a great deal of uncertainty in the durations involved. Research and development efforts and projects involving rapidly changing technology will likely call for a PERT approach. Houses, bridges, or other construction projects using standard materials have activity elements that can be estimated with some certainty based on past history and will normally utilize CPM. Construction of a dam, for example, would likely call for CPM; if, however, the construction were to be taking place in a developing nation, the availability of manpower, equipment, and materials may necessitate the PERT approach.

PERT calls for the user to assume that the uncertainty of each activity's time may be described by a probability distribution. The *most likely time* estimate would be the mode or highest value of the distribution. This is our estimate of the most frequently occurring number of days that would be required if the activity were performed many times under similar circumstances. The other two estimates are sometimes called the *optimistic time* and the *pessimistic time*. The optimistic activity time would be the shortest reasonable time for activity completion, that is, if nearly everything goes right. The pessimistic time would be the time required if Murphy's law² were encountered, that is, if most everything goes poorly. For highly uncertain activities, the time between estimates is likely to be large; for virtually certain activity durations, all three time estimates may be equal.

Before continuing with our discussion of PERT, we shall define some additional notation as follows:

- t_o = optimistic time
- t_m = most likely time
- t_p = pessimistic time
- t_e = expected (average) time

The first three times can be combined in a weighted average to determine an expected time. This weighted average is

$$t_e = \frac{t_o + 4t_m + t_p}{6} \quad (17.1)$$

Four times the weight is given to the most likely activity time than to the other two estimates. The expected time, t_e , is the average time the activity would take if it were repeated many times. This differs from t_m , which is the most frequently occurring duration in many repetitions of the activity.

Not only are we interested in an average activity duration, but we would also like a measure of the reliability of the estimate. Stated differently in a statistical

²A version of Murphy's law is as follows: "If there is even the most remote chance that something may go wrong, it will."

sense, we would like to know the variability or spread of the probability density function underlying the different times that may be taken to complete an activity. We recall that two standard measures of distribution spread are the variance and its square root, the standard deviation. The standard deviation may be estimated by the formula

$$S_t = \frac{t_p - t_o}{6} \quad (17.2)$$

This indicates that the standard deviation is $\frac{1}{6}$ of the spread between the maximum time and the minimum time. This is not a bad assumption in that we frequently consider virtually all of a unimodal (one peak) distribution to be contained within a spread of six standard deviations. Since the standard deviation is the square root of the variance of a distribution, the variance of t is given as follows:

$$V_t = \left(\frac{t_p - t_o}{6} \right)^2 \quad (17.3)$$

As an example of the PERT estimating procedure, suppose we determine that our three times for an activity are

$$t_o = 7, \quad t_m = 9, \quad t_p = 13$$

The expected or average time by the PERT formula is

$$t_e = \frac{t_o + 4t_m + t_p}{6} = \frac{7 + 4(9) + 13}{6} = 9.33 \quad (17.4)$$

The variance of the activity time distribution is found to be

$$V_t = \left(\frac{t_p - t_o}{6} \right)^2 = \left(\frac{13 - 7}{6} \right)^2 = 1$$

The standard deviation is therefore also equal to one. The distribution of time for this example is illustrated in Figure 17.7.

Now let us return to our example having activities A through I. But now let us assign PERT estimates instead of the single CPM activity duration. Table 17.2 indicates PERT estimates for each activity in our network. The three PERT activity

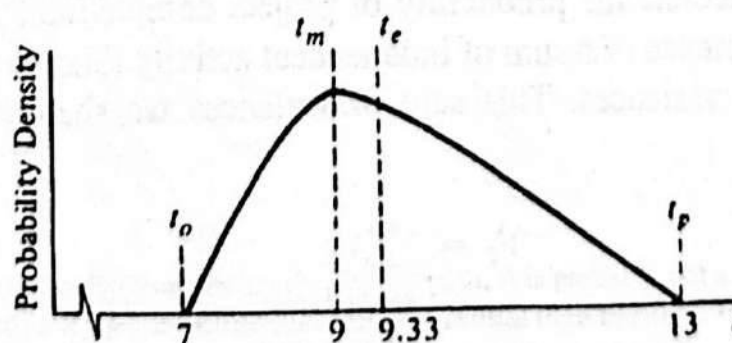


Fig. 17.7. Example activity time distribution.

Table 17.2. EXAMPLE PERT ACTIVITY TIME ESTIMATES

Activity	Optimistic Time, t_o	Most Likely Time, t_m	Pessimistic Time, t_p	Expected Time, t_e	Variance, V_i
A	5	7	15	8	2.778
B	14	21	22	20	1.778
C	21	33	45	33	16.000
D	14	16	30	18	5.444
E	18	19	26	20	1.778
F	7	9	11	9	.444
G	7	9	17	10	2.778
H	8	8	8	8	.000
I	3	4	5	4	.111

times must be selected based on the estimator's best judgment. In Table 17.2, however, we "handpicked" the times in such a way that the expected time for an activity, t_e , equals the CPM estimate for the same activity. In this way, we shall be able to better compare CPM and PERT through the same example.

In PERT the expected length of a sequence of activities, T_E , is equal to the sum of the respective activity expected times. That is,

$$T_E = \sum_{\substack{\text{all activities} \\ \text{along a path} \\ \text{of interest}}} t_e \quad (17.5)$$

This gives us an easy way of determining the length of any path through the network. In fact, it is identical to the method used for CPM! Therefore, the network representation, forward and backward pass calculations, total slacks, and critical path found for our CPM example are directly applicable to our PERT example.

With PERT, though, we can glean even more information from the network. If, in addition to knowing the mean or expected time along the critical path, we knew the variance along the critical path and the overall probability distribution of time, we could calculate the probability of project completion by a certain time. Fortunately, the variance of a sum of independent activity times is equal to the sum of their individual variances. This sum of variances we shall denote as V_T and write as

$$V_T = \sum_{\substack{\text{all activities} \\ \text{along a path} \\ \text{of interest}}} V_i \quad (17.6)$$

Naturally, the path standard deviation, S_T , is written as

$$S_T = \sqrt{V_T} \quad (17.7)$$

Further, from the central limit theorem we can deduce that the distribution of the sum of n independent activities approaches the normal distribution as n grows large. We already saw that the normal distribution is used in quality control work with subgroups as small as size 4 or 5. The typical project requiring a critical path approach will normally have at least this number of activities along its critical path.

Let us now calculate the mean and variance of the critical path in our example. Only two activities, C and G, comprise the critical path.³ The total expected time is

$$T_E = \sum_{C,G} t_e = 33 + 10 = 43$$

The variance along the critical path is

$$V_T = \sum_{C,G} V_i = 16.000 + 2.778 = 18.778$$

The standard deviation equals

$$S_T = \sqrt{V_T} = \sqrt{18.778} = 4.333$$

We are now ready to calculate the probability of project completion by, say, 47 days. Graphically, our problem is illustrated in Figure 17.8 in which a normal distribution having mean 43 and standard deviation 4.333 is shown. We want to know the area under the crosshatched portion of the normal curve. This represents the probability of project completion by time 47. First, we calculate the standard normal deviate, Z :

$$Z = \frac{X - \mu}{\sigma} = \frac{X - T_E}{S_T} = \frac{47 - 43}{4.333} = 0.923$$

Now, from the cumulative normal table in Appendix B, we find that the probability of a value less than $Z = 0.923$ is 0.822. In other words, there is an 82.2% chance of completing the critical path projects in 47 days. A similar calculation may be made for any number of days. Of course, the probability of completing the critical path by 43 days is 50%.

³With only two activities comprising the critical path, it is probably not a good idea to put much stock in the assumption of a normal distribution from the central limit theorem. However, the procedure will be continued for the sake of presenting an example of the method used.

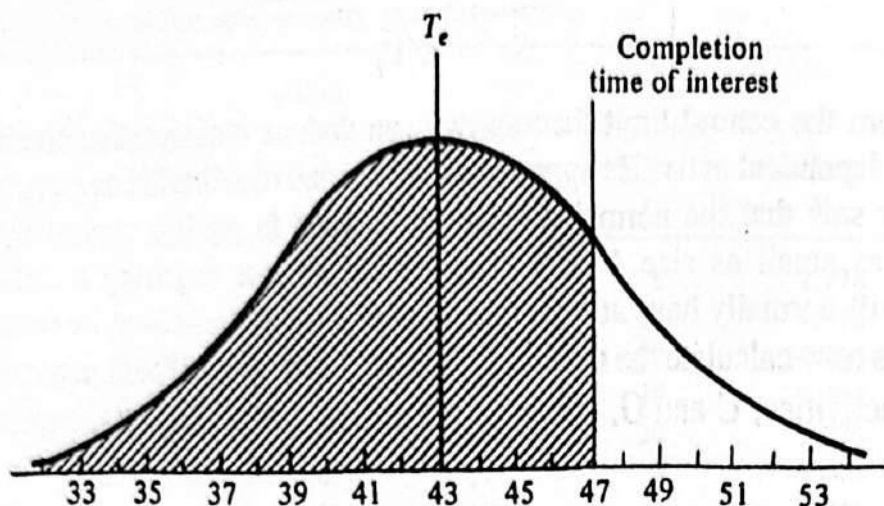


Fig. 17.8. Critical path probability distribution.

17.5. Time–Cost Trade-offs

In both CPM and PERT we have been using activity time estimates based on normal working conditions such as a regular work day or standard crew size. Almost all jobs, however, can be reduced in duration if additional resources are applied. Additional manpower, overtime, heavy equipment, and other resources increase the direct cost of an activity, however, and it only pays to use them if there is some incentive to do so. Often that incentive exists, and it may be present in many different forms. Indirect costs such as management, equipment, rentals, fixed expense allocation, and other “overhead” items normally increase as a function of project duration. Another form of indirect cost may be a contract performance clause providing for cost penalties for time overruns as well as monetary rewards for completion ahead of schedule. For example, it is not uncommon in the aerospace industry to have contract penalty or reward clauses of from \$50,000 to \$500,000 per day for late or early delivery, respectively.

We can see that project schedules influence both direct and indirect costs. Our objective as planners and schedulers is to trade off direct cost increases against indirect cost decreases, in search of an overall minimum total cost. We begin by estimating for each activity the absolute minimum activity completion time if all the necessary resources are available. This minimum time is called the *crash time*. A typical direct cost curve is illustrated in Figure 17.9. In this example the solid curved line may be our best estimate of direct cost vs. activity time; however, it is often sufficiently accurate to use a straight line between the crash and normal time-cost points. Here we have a slope of

$$\frac{\$9,000 - \$4,000}{10 \text{ days} - 5 \text{ days}} = \$1,000 \text{ per day}$$

which represents the direct cost of decreasing the activity time by 1 day. We assume that any integer day from 5 days through 10 days may be achieved. Returning to

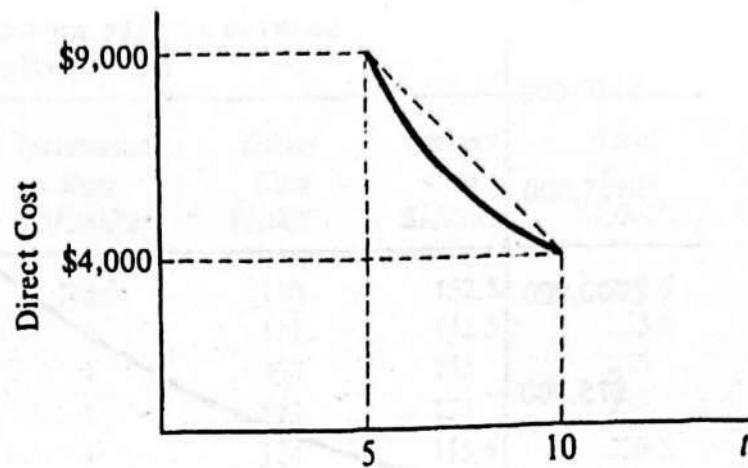


Fig. 17.9. Direct cost vs. activity duration.

our CPM example, we see that the normal and crash times and costs and the activity direct cost slopes are summarized in Table 17.3.

Once we have established normal and crash times, normal and crash costs, and direct cost slopes for each activity, we must establish the indirect costs as a function of project time. The total indirect cost over time for our project is shown in Figure 17.10. The indirect costs are also tabulated in Table 17.4. The indirect costs given include penalty and performance clauses, rentals, wages, and so on, as estimated from various contracts and financial records.

Having specified our direct and indirect costs as a function of time, we can begin systematically reducing activity times to see if a lower total project cost may be achieved. For example, in our problem the project duration using normal activity times is 43 days, as illustrated in Figure 17.6. If we can shorten the project duration from 43 days to 42 days, our indirect costs will drop from \$152,500 to \$142,500, according to Figure 17.10.

The critical path through the network consists of activities C and G. Either activity can be reduced by 1 day, dropping the project duration to 42 days. Of course, we do not want to randomly reduce the activity duration of either C or G. We want to choose the one that will increase direct costs by the least amount. From Table 17.3, we decide to decrease C by 1 day at a cost of \$1,000 instead of decreasing G by 1 day for \$3,000. Reducing C from 33 days to 32 days in duration increases total direct costs from \$110,000 to \$111,000, while reducing indirect costs from \$152,500 to \$142,500, providing a net savings of \$9,000.

Table 17.3. ACTIVITY NORMAL AND CRASH TIMES AND COSTS
(Costs are in thousands of dollars)

Activity	A	B	C	D	E	F	G	H	I
Normal time	8	20	33	18	20	9	10	8	4
Crash time	6	16	21	14	17	7	7	8	3
Normal cost	10	22	30	20	6	4	9	3	6
Crash cost	15	38	42	26	7.5	8	18	3	9.5
Slope (cost/day)	2.5	4	1	1.5	.5	2	3	—	3.5

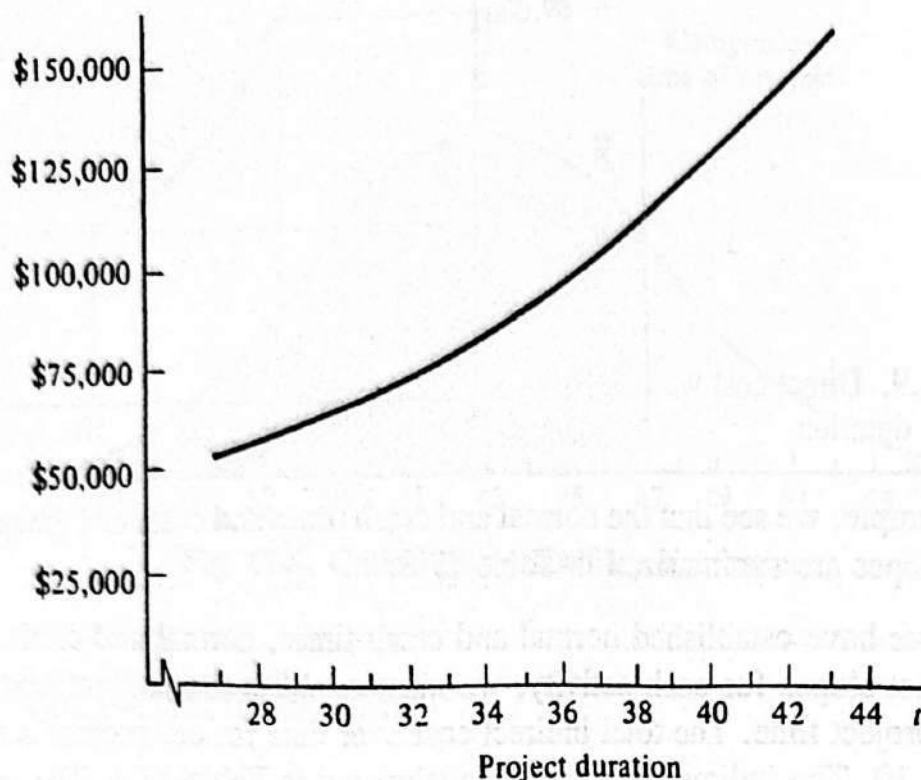


Fig. 17.10. Example indirect costs.

We now consider reducing the project duration further, and we analyze the network in the same way as before. Again, we decide to reduce activity C by another day. In fact, we continue reducing C day by day (from 32 days to 24 days), providing a project duration of 34 days with a direct cost of \$119,000 and an indirect cost of \$80,500. At a project duration of 34 days, illustrated in Figure 17.11, we have multiple critical paths. That is, both the C, G and A, D, H activity paths have durations of 34 days. Reducing the project length from this point in an optimal (least total project cost) manner is no longer a simple task.

There are several options open to us in which we can reduce project time from 34 days to 33 days. We can reduce each of the activities in any of the following pairs by 1 day each to achieve a project duration of 33 days: A, C; D, C; H, C; A, G; D, G; and H, G. Those pairs containing activity H can be ruled out immediately since we see from Table 17.3 that H cannot be reduced. Further, since C can be reduced even more at a direct cost of \$1,000 per day, we shall choose it over reducing activity G in the lower path. Finally, we decide to reduce activity D in the upper path at a cost of \$1,500 per day instead of shortening activity A at a cost of \$2,500 per day. The result of our deliberations is to reduce activity D from 18 days to 17 days and activity C from 24 days to 23 days, increasing direct cost from \$119,000 to \$121,500 and decreasing indirect cost from \$80,500 to \$75,000, for a net gain of \$3,000.

We have now achieved a project duration of 33 days and there are three critical paths: C, G; A, D, H; and B, F, I. In reducing the network duration further, we find that the direct cost of reducing the network tends to increase at a faster rate. This is because activities such as C with a low cost slope of \$1,000 per day

Table 17.4. TIME-COST TRADE-OFFS FOR EXAMPLE NETWORK
(Costs are in thousands of dollars)

<i>Project Duration</i>	<i>Activity Changes</i>	<i>Incremental Cost \$1,000's</i>	<i>Direct Cost \$1,000's</i>	<i>Indirect Cost \$1,000's</i>	<i>Total Cost \$1,000's</i>
43	None	None	110	152.5	262.5
42	C: 33 to 32	1	111	142.5	253.5
41	C: 32 to 31	1	112	133	245
40	C: 31 to 30	1	113	124	237
39	C: 30 to 29	1	114	115.5	229.5
38	C: 29 to 28	1	115	107.5	222.5
37	C: 28 to 27	1	116	100	216
36	C: 27 to 26	1	117	93	210
35	C: 26 to 25	1	118	86.5	204.5
34	C: 25 to 24	1	119	80.5	199.5
33	C: 24 to 23	1			
	D: 18 to 17	1.5	121.5	75	196.5
32	C: 23 to 22	1			
	D: 17 to 16	1.5			
	F: 9 to 8	2	126	70	196
31	C: 22 to 21	1			
	D: 16 to 15	1.5			
	E: 20 to 19	.5			
	F: 8 to 7	2	131	65.5	196.5
30	D: 15 to 14	1.5			
	G: 10 to 9	3			
	I: 4 to 3	3.5	139	61.5	200.5
29	A: 8 to 7	2.5			
	B: 20 to 19	4			
	G: 9 to 8	3	148.5	58	206.5
28	A: 7 to 6	2.5			
	B: 19 to 18	4			
	G: 8 to 7	3	158	55	213

eventually reach their crash time and cannot be reduced further and we must resort to shortening more costly activities such as G with a cost slope of \$3,000 per day. A complete summary of all project duration reductions and costs is given in Table 17.4.

The shortest project duration possible is estimated to be 28 days. That is, the critical path(s) cannot be reduced below 28 days since all the activities along at least one of the paths have been decreased to their crash times. For example, activity C is at 21 days and activity G is at 7 days. Neither activity may be reduced further. We should also note that the minimum total cost of \$96,000 occurs at a project length of 32 days. We have performed a "trade-off of costs," initially swapping increases in direct cost for larger decreases in indirect costs. This continued until at 31 days the increase in direct cost exceeded the decrease in indirect cost. Thus the

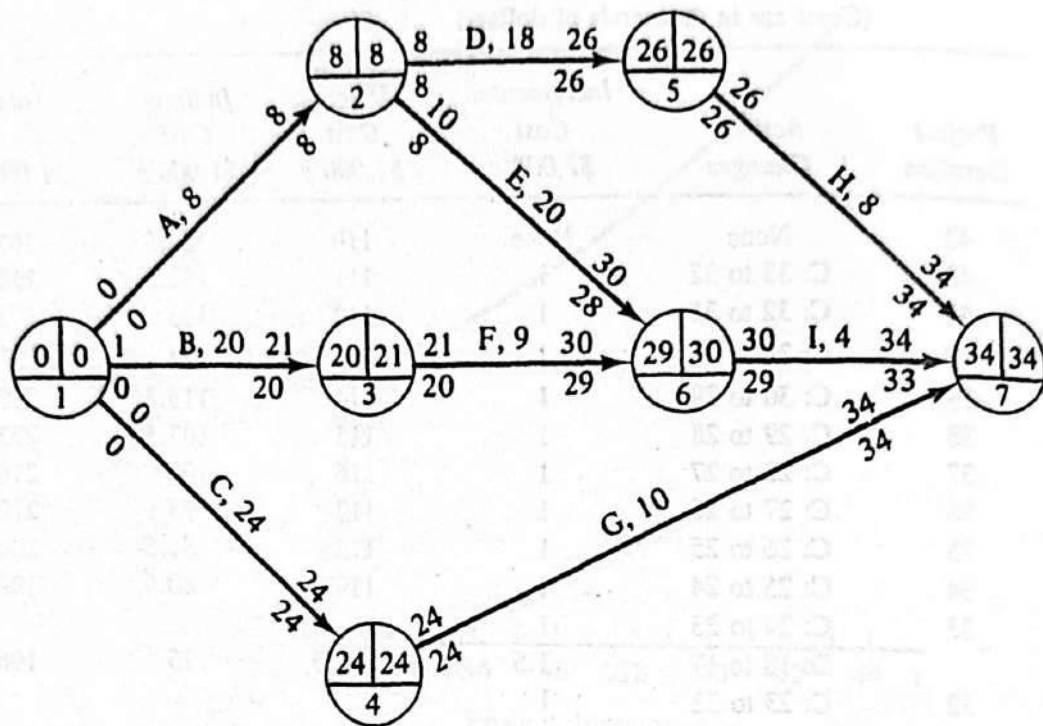


Fig. 17.11. Network compressed to 34 days.

optimum project length is 32 days. The concept of “trade-off of costs” is illustrated in Figure 17.12.

The procedure used for reducing the network in our example was largely one of complete enumeration and common sense. We can see, however, that for a network that has hundreds of activities, this approach would be infeasible. Numerous computer packages suitable for computer use on large projects have been developed.

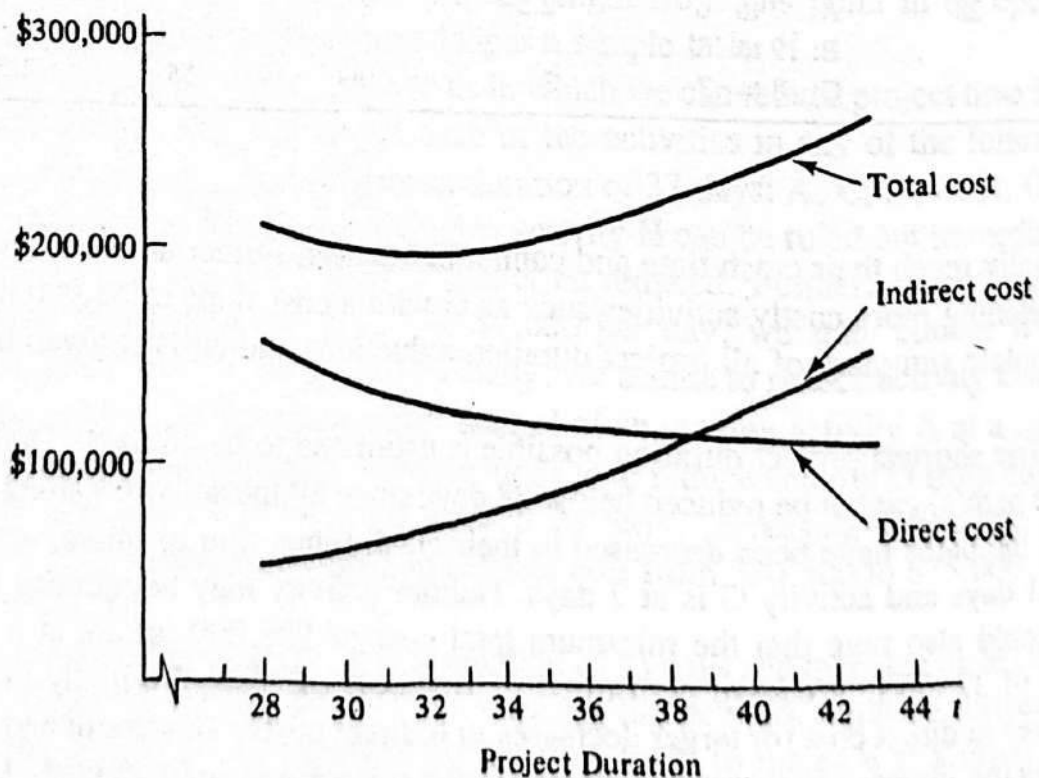


Fig. 17.12. Trade-off of costs in example network.

17.6. Resource Leveling

Often when we schedule the manpower resources for the activities of a project, some days require a large number of workers and other days require very few. The reason for this is that we have probably scheduled our manpower based on each activity beginning at its "earliest start" time. We should remember, however, that each activity may also begin as late as its "latest start" time. This flexibility, which we previously called slack, may be used to delay noncritical projects in such a way that fluctuations in manpower needed may be greatly reduced. That is, our resource needs over time may be leveled.

The obvious advantage of resource leveling is that we need not maintain a work force based on the highest manpower requirements originally scheduled. Further, through resource leveling we can minimize worker idle time. The procedures for resource leveling are beyond the scope of this book.